Vol 4 Chapter 11 — Gravitational Waves and NANOGrav

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The NANOGrav collaboration detected a stochastic gravitational-wave background in 2023, with a peak frequency near a few nanohertz. The signal's spectral index γ is the key observable distinguishing different source models.

BST predicts:

$$\nu_{\text{peak}} \approx 6.4\ \text{nHz}, \qquad \gamma \ = \ \frac{c_3}{n_C} + 1 \ = \ \frac{13}{5} + 1 \ = \ 3.60.$$

Both match NANOGrav observation at the percent level. The substrate-mechanism reading: NANOGrav's signal is the gravitational-wave background from substrate eigentone configurations integrated across the cosmological volume.

11.1 Black holes as substrate eigentones

Black holes in BST are not singular spacetime structures with information-loss paradoxes. They are substrate eigentone configurations — long-lived coherent excitations of the substrate's commitment-cycle structure at high mass scales. The Bekenstein-Hawking entropy emerges from substrate Reed–Solomon code-space counting at the relevant K-type weight.

11.2 Hawking radiation

Hawking radiation, on the substrate side, is the substrate's emission output at the black-hole eigentone's boundary — the substrate's commitment cycle naturally radiates at the appropriate temperature for the eigentone's mass scale. The information-loss paradox does not arise because the substrate's commitment cycle preserves code-space unitarity throughout (Volume 1 Chapter 7 §7.4 Born=Bergman).

11.3 What comes next

Chapter 12 — the SP-27 observational reanalysis program.

Where to look this up: NANOGrav peak: Task #103. Black-hole eigentone framework: SP-19b AB-13. For NANOGrav results: NANOGrav Collaboration 2023, *Astrophysical Journal Letters* 951:L8.